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WHEEL COVER WITH CUSTOMIZABLE BUTTON CAP

Abstract

A wheel cover for a wheel of a vehicle may include a cover portion removably operably coupleable to the wheel via a first set of retention tabs, the cover portion may be centered at an axis of rotation of the wheel and may cover spokes of the wheel, and a button cap removably operably coupleable to the wheel cover via a second set of retention tabs, the button cap may be centered at the axis of rotation of the wheel. The first set of retention tabs may be urged by a wire ring into engagement with a lip of the wheel, the lip may extend around the axis of rotation at a center bore of the wheel. The second set of retention tabs may insert into corresponding ones of a set of retention slots disposed in the cover portion.

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Background/Summary

TECHNICAL FIELD

[0001] Example embodiments generally relate to wheel covers and, more particularly, relate to a button cap for a wheel cover assembly.

BACKGROUND

[0002] Many modern vehicles, including battery electric vehicles (BEV), hybrid electric vehicles (HEV), and internal combustion engine vehicles (ICE), may wish to maximize efficiency and thus achieve a maximum possible operating range. To accomplish this goal, vehicles are being produced including increasingly lighter weight components, and more aerodynamic components, to reduce weight and drag coefficient of the vehicle. One example of a common component used among modern vehicles may be alloy wheels due to their relatively low weight and high durability compared to other types of wheels. To add to the aerodynamic performance of the alloy wheels, and thus reduce their drag coefficient, many vehicles may include the option of having wheel covers disposed on the wheels to provide a more aerodynamic design that generates less drag and increases the overall efficiency and operating range of the vehicle.

[0003] The wheel covers may sometimes attach to the wheel via a hub secure, and may also include a button cap often glued or snapped onto the wheel cover, which in many cases may rarely be removed or changed out for a different button cap. Due to the wheel covers being designed primarily to reduce drag, drivers of the vehicle may desire to change out or remove the wheel covers depending on the use case of the vehicle to balance looks with aerodynamic performance. In such cases, the button cap may provide drivers with a simple way to customize the vehicle on their own. Thus, rather than produce different button caps for different driving scenarios, there may be a desire to create a button cap that may be used with multiple different wheel covers and/or with the wheels without wheel covers.

BRIEF SUMMARY OF SOME EXAMPLES

[0004] In accordance with an example embodiment, a wheel cover for a wheel of a vehicle of an example embodiment may be provided. The wheel cover may include a cover portion removably operably coupleable to the wheel via a first set of retention tabs, the cover portion may be centered at an axis of rotation of the wheel and may cover spokes of the wheel, and a button cap removably operably coupleable to the wheel cover via a second set of retention tabs, the button cap may be centered at the axis of rotation of the wheel. The first set of retention tabs may be urged by a wire ring into engagement with a lip of the wheel, the lip may extend around the axis of rotation at a center bore of the wheel. The second set of retention tabs may insert into corresponding ones of a set of retention slots disposed in the cover portion.

[0005] In another example embodiment, a wheel assembly for a vehicle may therefore be provided. The wheel assembly may include a wheel which may be operably coupled to the vehicle, the wheel may include a lip that may extend around an axis of rotation at a center bore of the wheel, a wheel cover that may be removably operably coupleable to the lip via a first set of retention tabs, the wheel cover may be centered at the axis of rotation of the wheel to cover spokes of the wheel, and a button cap that may be removably operably coupleable to the wheel cover via a second set of retention tabs, the button cap may be centered at the axis of rotation of the wheel. The first set of retention tabs may be urged by a wire ring into engagement with the lip. The second set of retention tabs may insert into corresponding ones of a set of retention slots disposed in the wheel cover.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0006] Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0007] FIG. 1 illustrates a block diagram of a wheel assembly in accordance with an example embodiment;

[0008] FIG. 2 depicts a perspective view of the wheel assembly in accordance with an example embodiment;

[0009] FIG. 3 depicts an exploded perspective view of the wheel assembly in accordance with an example embodiment;

[0010] FIG. 4 illustrates a perspective view of the interior surface of the cover portion in accordance with an example embodiment;

[0011] FIG. 5 illustrates a close up perspective view of the first set of retention tabs taken from box 5 in FIG. 4 in accordance with an example embodiment;

[0012] FIG. 6 illustrates a close up perspective view of the button cap and the recessed portion in accordance with an example embodiment;

[0013] FIG. 7 illustrates a top view of the wheel cover with the button cap removed in accordance with an example embodiment;

[0014] FIG. 8 illustrates a close up top view of the recessed portion taken from box 8 in FIG. 7 in accordance with an example embodiment;

[0015] FIG. 9 illustrates an isolated top side perspective view of the button cap in accordance with an example embodiment;

[0016] FIG. 10 illustrates an isolated bottom side perspective view of the button cap in accordance with an example embodiment;

[0017] FIG. 11 illustrates a section view of the wheel assembly in accordance with an example embodiment;

[0018] FIG. 12 illustrates a section view of the button cap operably coupled to the wheel cover in accordance with an example embodiment;

[0019] FIG. 13 illustrates a section view of the button cap operably coupled to the wheel without the wheel cover in accordance with an example embodiment; and

[0020] FIG. 14 illustrates a perspective view of the button cap operably coupled to the wheel without the wheel cover in accordance with an example embodiment.

DETAILED DESCRIPTION

[0021] Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed, the examples described and pictured herein should not be construed as being limiting as to the scope, applicability or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable requirements. Like reference numerals refer to like elements throughout. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other.

[0022] Some example embodiments described herein may address the problems described above. In this regard, for example, some embodiments may provide a wheel cover for a wheel of a vehicle that may include a customizable button cap. The button cap of some embodiments may be interchangeably operably coupled to the wheel cover and to the wheel alone without the wheel cover. Thus the wheel and the wheel cover may be customized by changing out the button cap or changing out a graphic disposed inside of the button cap. Additionally, the wheel cover and the button cap may be more robustly operably coupled to the wheel, and thus the wheel cover and button cap may be less likely to be inadvertently removed from the wheel.

[0023] FIG. 1 illustrates a block diagram of a wheel assembly **100** for a vehicle according to an example embodiment, FIG. 2 depicts a perspective view of the wheel assembly **100** of an example embodiment, and FIG. 3 depicts an exploded view of the wheel assembly **100** in accordance with an example embodiment. Referring now to FIGS. 1-3, the wheel assembly **100** may include a wheel **110**, a wheel cover **120** and a button cap **130**. The wheel **110** may be operably coupled to the vehicle via a wheel hub (not shown) disposed at the vehicle. The wheel hub may be operably coupled to the wheel **110** at a center bore **114** of the wheel **110**, which may be a circular orifice that may include an axis of rotation **116** extending perpendicularly through the center. The wheel **110** may also include a lip **112** that may extend around the center bore **114** of the wheel **110**. The lip **112** may also extend around the axis of rotation **116** of the wheel **110**, which may also extend through the center of the wheel cover **120** and the button cap **130**.

[0024] In some cases, the lip **112** may be disposed radially inward of spokes **118** of the wheel **110**, which may extend from the center bore **114** out towards a perimeter of the wheel **110**. The wheel cover **120** may be centered at the axis of rotation **116** to cover an outer face of the wheel **110** while the wheel cover **120** may be operably coupled to the wheel **110**. The wheel cover **120** may be removably operably coupleable to the wheel **110** via a first set of retention tabs **140** and a set of clips **160**. The first set of retention tabs **140** and the set of clips **160** will be discussed in more detail below in reference to FIGS. 4 and 5. The button cap **130** may be removably operably coupleable to the wheel cover **120** via a second set of retention tabs **150** in a position such that the button cap **130** may be centered at the axis of rotation **116** of the wheel **110**. The wheel cover **120** of some example embodiments may include a cover portion **122**, and the cover portion **122** may include an interior surface **124** that may face inwardly toward the wheel **110** and an exterior surface **126** that may face outwardly away from the wheel **110** when the wheel cover **120** may be operably coupled to the wheel **110**. The cover portion **122** may provide aerodynamic design enhancements to the wheel **110** (i.e. a reduced drag coefficient of the wheel **110**) by covering the spokes **118** and redirecting airflow to flow by the wheel **110** smoother when the vehicle is driven. The first set of retention tabs **140** may therefore operably couple the wheel cover **120** to an outer face of the wheel **110** such that the spokes **118** may be covered by the cover portion **122**, as shown in FIG. 2.

[0025] FIG. 4 depicts a perspective view of the interior surface **124** of the wheel cover **120** according to an example embodiment, and FIG. 5 illustrates a close up perspective view of the first set of retention tabs **140** taken from box 5 in FIG. 4. In some cases, the first set of retention tabs **140** may include a wire ring **142** which may urge the first set of retention tabs **140** into engagement with the lip **112** to removably operably couple the wheel cover **120** to the wheel **110**. In some cases, the wheel cover **120** may be formed from a plastic material, a polymer material, a composite material, or any combination thereof, perhaps through an injection molding process. As such, the first set of retention tabs **140** may be formed from the same material as the cover portion **122**, and perhaps formed during the same process as the cover portion **122** as well. The first set of retention tabs **140** may be arranged in a circle pattern so that individual ones of the first tabs **144** in the first set of retention tabs **140** may operably couple to a different portion of the lip **112**. In this regard, the first tabs **144** may therefore be radially disposed and equidistantly spaced apart from one another. This may allow the first set of retention tabs **140** to evenly distribute forces generated while the cover portion **122** may be operably coupled to the wheel **110** amongst the individual first tabs **144** that make up the first set of retention tabs **140**. In an example embodiment, the first set of retention tabs **140** may include five individual first tabs **144**. In some other cases, more or less may be used depending on the size of the wheel **110**.

[0026] Also shown in FIG. 4, the set of clips **160** may include a plurality of individual clips **162** that may operably couple the cover portion **122** to the spokes **118** of the wheel **110**. In this regard, the set of clips **160** may be disposed radially outward of the first set of retention tabs **140** on the interior surface **124** of the cover portion **122**. In some cases, the clips **162** may be disposed in pairs, and each pair of clips **162** may operably couple to a single spoke **118** of the wheel **110**. In this

regard, in each pair of clips **162**, one clip **162** may operably couple to a first side of the spoke **118** and the other clip **162** in the pair may operably couple to a second side of the spoke **118**. As such, each spoke **118** may be operably coupled to a clip **162** on each side of the spoke **118**, and the set of clips **160** may therefore provide additional support for the wheel cover **120** relative to the wheel **110**. In other words, the set of clips **160** may prevent the cover portion **122** from rattling against the spokes **118** and creating noises and vibrations. Thus, together, the first set of retention tabs **140** and the set of clips **160** may operably couple the wheel cover **120** to the wheel **110**.

[0027] As seen in FIG. 5, the wire ring **142** may be formed from a metallic material disposed in a C shape to provide a biasing force to the first tabs **144**. In this regard, the wire ring **142** may be disposed on a radially interior side of the first set of retention tabs **140** and may urge each first tab **144** radially outward and into engagement with the lip **112**. In an example embodiment, each first tab **144** may include a wire holder **146** disposed on a radially interior surface of each first tab **144**. The wire holder **146** may operably couple the wire ring **142** to each first tab **144** so that the wire ring **142** may not move out of position relative to the first set of retention tabs **140**. In some cases, the wire ring **142** may operably couple to each wire holder **146** via a snap fit. In an example embodiment, the wire ring **142** may not be a complete closed circle ring shape. In other words, the wire ring **142** may instead be a C shaped member. Functionally similar to an internal C-clip or retaining ring, the wire ring **142** may provide a biasing force that may “open” the C shape wider. This force may urge the first tabs **144** radially outward to engage the first tabs **144** with the lip **112**, but simultaneously, this force may also help hold the wire ring **142** in the wire holder **146** as well. In an example embodiment, the wire holder **146** may include upper and lower anchors that may operably couple to the wire ring **142** via the snap fit. In this regard, the wire ring **142** may snap into the wire holder **146** in between the upper and lower anchors.

[0028] The first set of retention tabs **140** may also include a first projection **148** and a first set of spacer members **170** disposed on a radially exterior side of each first tab **144** of the first set of retention tabs **140**. In this regard, the radially exterior side of each first tab **144** may be disposed opposite from the radially interior side of each first tab **144**. For example, the radially interior side of the first tab **144** may face inwardly with respect to the circular orientation of the first set of retention tabs **140**, the radially exterior side of the first tab **144** may face outwardly with respect to the circular orientation of the first set of retention tabs **140**. The first projection **148** may be disposed at a distal end of each first tab **144**, away from where the first tab **144** operably couples to the cover portion **122**. In some cases, as the wheel cover **120** is operably coupled to the wheel **110**, each instance of the first projection **148** may operably couple its respective first tab **144** to the lip **112** by first sliding beyond the lip **112** and then engaging a back side of the lip **112** by snapping in place due to the bias force provided by the wire ring **142**. In this regard, the first projection **148** may be disposed at the back side of the lip **112** while the wheel cover **120** may be operably coupled to the wheel **110**. On the other hand, the first set of spacer members **170** may be disposed closer to the cover portion **122** on the first tabs **144**, and spaced apart from the first projection **148**. The first set of spacer members **170** may be disposed at a front side of the lip **112** while the wheel cover **120** may be operably coupled to the wheel **110** such that the lip **112** may be disposed between the first projection **148** and the first set of spacer members **170**. In some cases, the first set of spacer members **170** may define a first standoff height (H1) between the wheel cover **120** and the wheel **110**. The engagement of each first tab **144** with the lip **112** will be described in further detail below in reference to FIG. 11.

[0029] Also seen in FIG. 5, the cover portion **122** may include a set of retention slots **180**. The set of retention slots **180** may include individual retention slots **182** that may be disposed in a circular pattern, similar to the first set of retention tabs **140**. Each retention slot **182** in the set of retention slots **180** may define an opening extending entirely through the cover portion **122**. Each slot **182** may be substantially elliptical in shape and may be arcuate as well. In this regard, the set of retention slots **180** may be disposed in a circular pattern around the perimeter of a recessed portion

128 at the center of the wheel cover **120**. In some cases, each slot **182** of the set of retention slots **180** may be disposed between consecutive first tabs **144** of the first set of retention tabs **140**. Thus, at the interior surface **124** of the cover portion **122**, the first tabs **144** and the slots **182** may alternate in the circular pattern. In other words, each first tab **144** may be disposed between two slots **182** and each slot **182** may be disposed between two first tabs **144**. In an example embodiment, the set of retention slots **180** may include five individual slots **182**. In some other cases, more or less may be used depending on the size of the wheel **110**.

[0030] FIG. **6** depicts the button cap **130** operably coupling to the recessed portion **128** of the wheel cover **120** according to an example embodiment. FIG. **7** illustrates a top perspective view of the wheel cover **120** showing the exterior surface **126** and the set of retention slots **180** in accordance with an example embodiment. FIG. **8** shows a close up top perspective view of the recessed portion **128** including the set of retention slots **180** taken from box **8** in FIG. **7**. FIGS. **9** and **10** show top and bottom perspective views of the button cap **130** according to an example embodiment. Referring now to FIGS. **6-10**, in some cases, individual second tabs **152** of the second set of retention tabs **150** may insert into corresponding slots **182** of the set of retention slots **180** to removably operably couple the button cap **130** to the wheel cover **120**. However, in other cases, the button cap **130** may be alternately operably coupleable to the wheel **110** without the wheel cover **120** and to the wheel cover **120** itself. In this regard, the wheel cover **120** may be removed from the wheel **110**, and the button cap **130** may be operably coupleable to the wheel **110** via the lip **112**, excluding the wheel cover **120**. This will be discussed in further detail later in reference to FIGS. **13** and **14**.

[0031] The button cap **130** may operably couple to the wheel cover **120** via the second set of retention tabs **150** inserting into the set of retention slots **180** and operably coupling thereto via a snap fit. In this regard, each second tab **152** of the second set of retention tabs **150** may include a second projection **158**. Each second projection **158** may pass through a respective retention slot **182** as the button cap **130** is inserted into the recessed portion **128**. The second projections **158** snap into place responsive to passing through the slot **182** to operably couple the button cap **130** to the wheel cover **120**. In such cases, the second tabs **152** of the second set of retention tabs **150** may engage the interior surface **124** of the cover portion **122** to hold the button cap **130** in place in the recessed portion **128**. The second set of retention tabs **150** may be arranged in a circle shape so that individual ones of the second tabs **152** in the second set of retention tabs **150** may operably couple to respective different slots **182** of the set of retention slots **180**. In this regard, the second tabs **152** may be radially disposed and equidistantly spaced apart from one another. This may allow the second set of retention tabs **150** to evenly distribute forces generated while the cover portion **122** may be operably coupled to the wheel **110** amongst the individual second tabs **152** that make up the second set of retention tabs **150**. In an example embodiment, the second set of retention tabs **150** may include five individual second tabs **152**. In some other cases, more or less may be used depending on the size of the wheel **110**.

[0032] Due to their snap fit operable coupling, the second tabs **152** may bend radially in and radially out responsive to the second projection **158** passing through the slot **182**. Over time, due to this repeated bending of the second tabs **152**, the second tabs **152** may experience material fatigue that may reduce the longevity of the second set of retention tabs **150** due to repeatedly inserting and removing the button cap **130** numerous times over its lifespan. In order to increase the resistance of the second tabs **152** to material fatigue, and increase the longevity of the second tabs **152**, the second tabs **152** may be formed through a dual shot molding process (also known as two shot injection molding, double shot injection molding, multi shot injection molding, etc.). Dual shot molding may allow two different materials to be included in a same injection molded component so that the component may exhibit desired characteristics of both materials. For example, the second tabs **152** may be dual shot molded to include a more flexible material disposed where the second tabs **152** meet the body of the button cap **130**, and to include a harder material disposed at the distal

end of the second tabs **152** where the second projection **158** may be disposed and where the second tabs **152** may engage the wheel cover **120**. In this regard, the more flexible material may permit the second tabs **152** to repeatedly bend with greater ease as the second projection **158** passes through the slot **182**, and the harder material may hold up better to engaging and disengaging with the wheel cover **120**, or with the lip **112**, repeatedly over time. As compared to the harder material, the more flexible material may last for many more cycles of the second tabs **152** being bent, which may increase the longevity of the second set of retention tabs **150**.

[0033] In some cases, the button cap **130** may include a second set of spacer members **190** that may define a second standoff height (H2) between the button cap **130** and the wheel cover **120**. In an example embodiment, individual second spacer members **192** of the second set of spacer members **190** may be disposed between consecutive second tabs **152** of the second set of retention tabs **150**. The second spacer members **192** may contact the exterior surface **126** of the cover portion **122** responsive to the second set of retention tabs **150** being inserted into the set of retention slots **180**. In some other cases, the second spacer members **192** may contact the lip **112** of the wheel **110** responsive to the button cap **130** being inserted into the center bore **114**. In an example embodiment, the first standoff height (H1) may be approximately equal to the second standoff height (H2). This may facilitate the button cap **130** operably coupling to the wheel cover **120** or directly operably coupling to the wheel **110** by ensuring that the second set of retention tabs **150** may engage either the lip **112** or the set of retention slots **180** at the same standoff height. In some cases, each second spacer member **192** of the second set of spacer members **190** may be disposed between consecutive second tabs **152** of the second set of retention tabs **150**. Thus, at an inside face of the button cap **130**, the second tabs **152** and the second spacer members **192** may alternate in a circular pattern around the perimeter of the button cap **130**. In other words, each second tab **152** may be disposed between two second spacer members **192** and each second spacer member **192** may be disposed between two second tabs **152**.

[0034] FIG. **11** depicts a section view of the wheel assembly **100** including the wheel **110**, the wheel cover **120**, and the button cap **130** according to an example embodiment. FIG. **12** illustrates a section view of the wheel cover **120** and the button cap, with the wheel **110** removed for visibility, in accordance with an example embodiment. FIG. **13** illustrates a section view of the button cap **130** and the wheel **110** of an example embodiment. Referring now to FIGS. **11-13**, the second standoff height (H2) may define a customization cavity **200** disposed between the button cap **130** and the wheel cover **120** in the recessed portion **128** of the cover portion **122**. The customization cavity **200** may be bound by the cover portion **122** and the button cap **130**, and may be customizable with a graphic **210**. In this regard, in some cases the button cap **130** may be formed from a translucent material that may allow light to pass through the button cap **130** and into the customization cavity **200**, and thus onto the graphic **210** (which may be better shown in FIGS. **6-8**). For example, the button cap **130** may be formed from a clear acrylic material, or any other similarly translucent material. In some cases, the graphic **210** may be a sticker having an emblem, logo, name, personal identifier, family crest, or any other visual marker as desired by the driver. In an example embodiment, the driver may create their own graphic **210** to display in the customization cavity **200**. In some cases, the driver may be able to purchase the graphic **210** that they wish to display in the customization cavity **200**. In an example embodiment, the second standoff height (H2) may be greater than the first standoff height (H1) to create a 3D effect with the graphic **210** within the customization cavity **200**. In such cases, the driver may be able to 3D print the graphic **210** and adhere it to the cover portion **122** due to the increased second standoff height (H2) providing more space in the customization cavity **200**.

[0035] In some other cases, the graphic **210** may include a photoluminescent material. In this regard, with the button cap **130** being formed from a translucent material, the photoluminescent material may “charge” during the day or in brighter areas, and may illuminate at night or in darker areas. In an example embodiment, some translucent materials may allow more light to pass through

than others. For instance, clear acrylic may be more efficient for allowing the graphic **210** comprising the photoluminescent material to charge than clear propylene, for example. This may be because clear propylene may only pass blue light through, which may greatly reduce the efficiency of charging the photoluminescent material. In an example embodiment, the button cap **130** itself may be molded in the photoluminescent material. In this regard, the button cap **130** may charge up during the day or in brighter areas, and may illuminate at night or in darker areas. In some cases, the photoluminescent material may be a long-persistent phosphor material. The long-persistent phosphor may be included in either the graphic **210** or the button cap **130** itself. In an example embodiment, the button cap **130** may be molded from the translucent material while the second set of retention tabs **150** may still be dual shot molded for the increased resistance to material fatigue. In some other cases, the graphic **210** may be dual shot molded into the recessed portion **128** of the wheel cover **120** in a different color material so that the graphic **210** may be visible in the customization cavity **200**.

[0036] FIG. **11** also shows the operable coupling of the first set of retention tabs **140** to the lip **112** of the wheel **110**. In FIG. **11**, the section through the wheel **110** is shown by the darkest shading, the section through the wheel cover **120** is shown by the medium shading, and the section through the button cap **130** is shown by the lightest shading. In this regard, as the wheel cover **120** may be inserted onto the wheel **110**, the first set of retention tabs **140** may engage the lip **112**. The first tabs **144** may extend beyond the lip **112** such that the first projection **148** may slide by the lip **112** as the wheel cover **120** is inserted. Responsive to the wheel cover **120** being inserted onto the wheel **110**, the first projection **148** and the first set of spacer members **170** may engage the lip **112**, due to being biased by the wire ring **142**, such that the lip **112** may be disposed between the first projection **148** and the first set of spacer members **170** at each first tab **144**. Thus, while the wheel cover **120** is operably coupled to the wheel **110**, the first set of spacer members **170** may define the first standoff height (H1) between the wheel cover **120** and the wheel **110**. Simultaneously, the button cap **130** may be operably coupled to the wheel cover **120** via the second set of retention tabs **150** and the set of retention slots **180**. The second set of spacer members **190** disposed at the button cap **130** may define the second standoff height (H2) of the customization cavity **200** by contacting an edge of the recessed portion **128**. As shown in FIG. **11**, the diameter of the circle pattern of the first set of retention tabs **140**, the second set of retention tabs **150**, and the set of retention slots **180** may all be the same. This may allow the button cap **130** to be installed on the wheel **110** alone (without the wheel cover **120**) or onto the wheel cover **120**, interchangeably.

[0037] FIG. **12** shows an alternate section view with the wheel **110** removed for added visibility. In FIG. **12**, the darkest shading represents the section through the wheel cover **120** while the lighter shading represents the section through the button cap **130**. The view of FIG. **12** may better show the second set of retention tabs **150** extending through the set of retention slots **180**, as well as the customization cavity **200**.

[0038] FIGS. **13** and **14** depict the button cap **130** operably coupled to the wheel **110** without the wheel cover **120**. As described above, because the first standoff height (H1) may be equal to the second standoff height (H2), and because the diameter of the circle pattern of the first set of retention tabs **140**, the second set of retention tabs **150**, and the set of retention slots **180** may all be the same, the button cap **130** may be alternately, and interchangeably, operably coupled to the wheel cover **120** and to the naked wheel **110**. In the example embodiment shown in FIG. **13**, the second set of spacer members **190** may contact the lip **112** to define the second standoff distance (H2) of the button cap **130** while the second set of retention tabs **150** engage the lip **112**. This may reduce the likelihood of the button cap **130** rattling in either the wheel **110** or the wheel cover **120** by not having the button cap **130** extend to an incorrect depth.

[0039] A wheel cover for a wheel of a vehicle of an example embodiment may therefore be provided. The wheel cover may include a cover portion removably operably coupleable to the wheel via a first set of retention tabs, the cover portion may be centered at an axis of rotation of the

wheel and may cover spokes of the wheel, and a button cap removably operably coupleable to the wheel cover via a second set of retention tabs, the button cap may be centered at the axis of rotation of the wheel. The first set of retention tabs may be urged by a wire ring into engagement with a lip of the wheel, the lip may extend around the axis of rotation at a center bore of the wheel. The second set of retention tabs may insert into corresponding ones of a set of retention slots disposed in the cover portion.

[0040] The wheel cover of some embodiments may include additional features, modifications, augmentations and/or the like to achieve further objectives or enhance performance of the wheel cover. The additional features, modifications, augmentations and/or the like may be added in any combination with each other. Below is a list of various additional features, modifications, and augmentations that can each be added individually or in any combination with each other. For example, the button cap may be alternately operably coupleable to the cover portion itself and to the wheel without the wheel cover. In an example embodiment, individual ones of the set of retention slots may be disposed between consecutive ones of the first set of retention tabs. In some cases, the wheel cover may include a first set of spacer members that may define a first standoff height between the wheel cover and the wheel. In an example embodiment, the first set of spacer members may be disposed at the first set of retention tabs. In some cases, the button cap may include a second set of spacer members that may define a second standoff height between the button cap and the cover portion. In an example embodiment, individual ones of the second set of spacer members may be disposed between consecutive ones of the second set of retention tabs. In some cases, the first standoff height may be equal to the second standoff height. In an example embodiment, the second standoff height may define a customization cavity that may be disposed between the button cap and the wheel cover, the customization cavity may be customizable with a graphic. In some cases, the button cap may be formed from a translucent material that may permit light to pass through the button cap and onto the graphic. In an example embodiment, the graphic may include a photoluminescent material disposed at the wheel cover beneath the button cap. In some cases, the second set of retention tabs may be dual shot molded to increase longevity and fatigue resistance. In an example embodiment, the wire ring may be disposed on a radially interior side of the first set of retention tabs and may urge the first set of retention tabs radially outward toward the lip.

[0041] A wheel assembly for a vehicle may therefore be provided. The wheel assembly may include a wheel which may be operably coupled to the vehicle, the wheel may include a lip that may extend around an axis of rotation at a center bore of the wheel, a wheel cover that may be removably operably coupleable to the lip via a first set of retention tabs, the wheel cover may be centered at the axis of rotation of the wheel to cover spokes of the wheel, and a button cap that may be removably operably coupleable to the wheel cover via a second set of retention tabs, the button cap may be centered at the axis of rotation of the wheel. The first set of retention tabs may be urged by a wire ring into engagement with the lip. The second set of retention tabs may insert into corresponding ones of a set of retention slots disposed in the wheel cover.

[0042] Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases

where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A wheel cover for a wheel of a vehicle, the wheel cover comprising: a cover portion removably operably coupleable to the wheel via a first set of retention tabs, the cover portion being centered at an axis of rotation of the wheel to cover spokes of the wheel; and a button cap removably operably coupleable to the wheel cover via a second set of retention tabs, the button cap being centered at the axis of rotation of the wheel, wherein the first set of retention tabs is urged by a wire ring into engagement with a lip of the wheel, the lip extending around the axis of rotation at a center bore of the wheel, and wherein the second set of retention tabs inserts into corresponding ones of a set of retention slots disposed in the cover portion.
2. The wheel cover of claim 1, wherein the button cap is alternately operably coupleable to the cover portion itself and to the wheel without the wheel cover.
3. The wheel cover of claim 1, wherein individual ones of the set of retention slots are disposed between consecutive ones of the first set of retention tabs.
4. The wheel cover of claim 1, wherein the wheel cover comprises a first set of spacer members that defines a first standoff height between the wheel cover and the wheel, and wherein the first set of spacer members is disposed at the first set of retention tabs.
5. The wheel cover of claim 4, wherein the button cap comprises a second set of spacer members that defines a second standoff height between the button cap and the cover portion, and wherein individual ones of the second set of spacer members are disposed between consecutive ones of the second set of retention tabs.
6. The wheel cover of claim 5, wherein the first standoff height is equal to the second standoff height.
7. The wheel cover of claim 5, wherein the second standoff height defines a customization cavity disposed between the button cap and the wheel cover, the customization cavity being customizable with a graphic.
8. The wheel cover of claim 7, wherein the button cap is formed from a translucent material to permit light to pass through the button cap and onto the graphic.
9. The wheel cover of claim 8, wherein the graphic comprises a photoluminescent material disposed at the wheel cover beneath the button cap.
10. The wheel cover of claim 1, wherein the second set of retention tabs are dual shot molded to increase longevity and fatigue resistance.
11. The wheel cover of claim 1, wherein the wire ring is disposed on a radially interior side of the first set of retention tabs to urge the first set of retention tabs radially outward toward the lip.
12. A wheel assembly for a vehicle, the wheel assembly comprising: a wheel operably coupled to the vehicle, the wheel comprising a lip that extends around an axis of rotation at a center bore of the wheel, a wheel cover removably operably coupleable to the lip via a first set of retention tabs, the wheel cover being centered at the axis of rotation of the wheel to cover spokes of the wheel; and a button cap removably operably coupleable to the wheel cover via a second set of retention tabs, the button cap being centered at the axis of rotation of the wheel, wherein the first set of retention tabs is urged by a wire ring into engagement with the lip, and wherein the second set of retention tabs inserts into corresponding ones of a set of retention slots disposed in the wheel cover.
13. The wheel assembly of claim 12, wherein the button cap is alternately operably coupleable to

the wheel without the wheel cover and to the wheel cover itself.

14. The wheel assembly of claim 12, wherein the wheel cover comprises a first set of spacer members that defines a first standoff height between the wheel cover and the wheel, and wherein the first set of spacer members is disposed at the first set of retention tabs.

15. The wheel assembly of claim 14, wherein the button cap comprises a second set of spacer members that defines a second standoff height between the button cap and the wheel cover, and wherein individual ones of the second set of spacer members are disposed between consecutive ones of the second set of retention tabs.

16. The wheel assembly of claim 15, wherein the first standoff height is equal to the second standoff height.

17. The wheel assembly of claim 15, wherein the second standoff height defines a customization cavity disposed between the button cap and the wheel cover, the customization cavity being customizable with a graphic.

18. The wheel assembly of claim 17, wherein the button cap is formed from a translucent material to permit light to pass through the button cap and onto the graphic.

19. The wheel assembly of claim 18, wherein the graphic comprises a photoluminescent material disposed at the wheel cover beneath the button cap.

20. The wheel assembly of claim 12, wherein the second set of retention tabs are dual shot molded to increase longevity and fatigue resistance.
